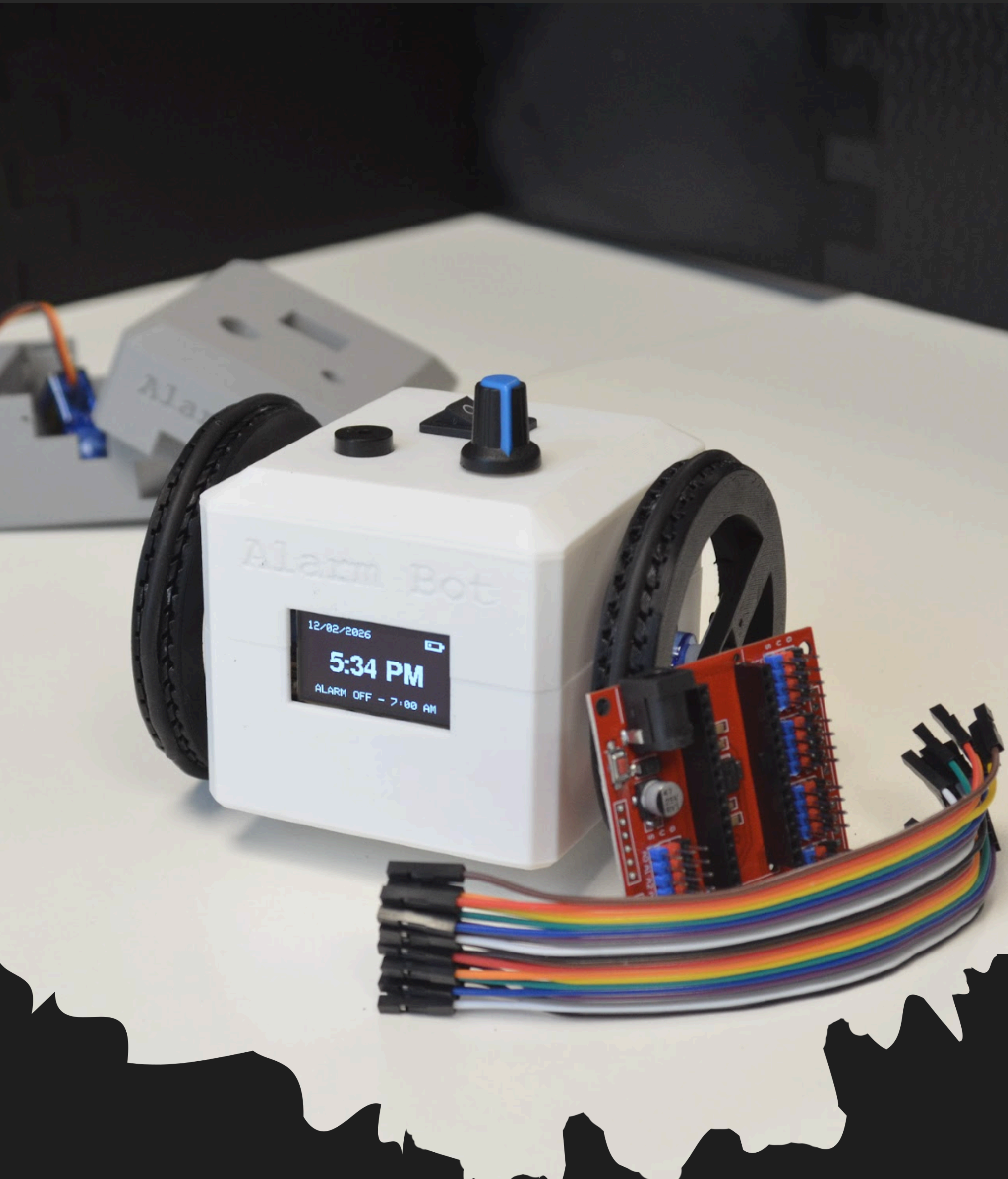


ALARM BOT WORKSHOP

INFORMATION PACK

Find out more at:
projectbeacon.org.au

ABOUT US

Project Beacon is a not-for-profit organisation based in Sydney whose goal is to bring the world of STEM directly into the hands of young student innovators. All our presenters are current university students pursuing STEM degrees, allowing them to provide an engaging and relatable learning experience for each student. Our interactive and curriculum-aligned workshops aim to teach students the fundamentals of robotics, from manufacturing through to programming and testing. Each workshop is built around students creating robots they can take home, inspiring continual experimentation and future learning.

WHAT IS AN ALARM BOT?

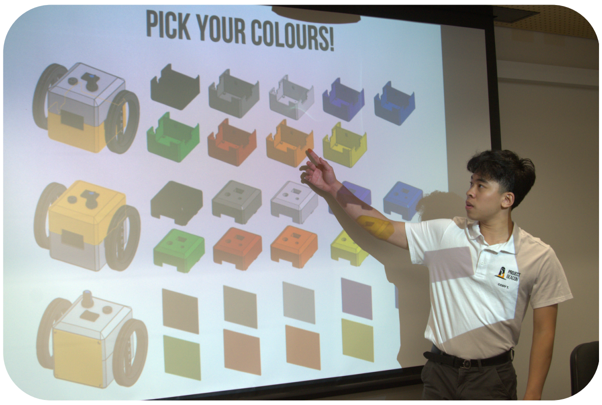
The Alarm Bot is a robot designed and 3D printed in-house by our team. While it includes all the similar functions of a typical alarm clock - a screen and buzzer - it also features two motorised wheels to drive across the room, preventing a heavy sleeper from pressing snooze and dozing back off. We directly chose this bot design due to its strong relevance to a student's everyday life, demonstrating how STEM can be integrated into our daily routines.

HOW WILL THE WORKSHOP RUN?

The workshop will run for approximately 2 hours and will be scaffolded into the following main sections:

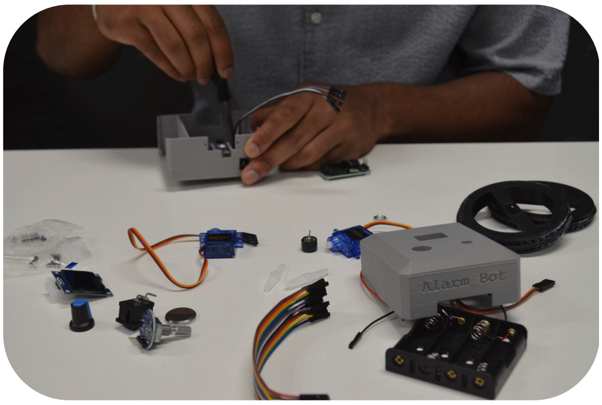
1. Onboarding Presentation (20 Minutes)

Students are introduced to the fundamentals of 3D printing and its unique advantages that made it the ideal manufacturing process for the Alarm Bot's chassis. They are then guided through the inner workings of the bot and how each of its electrical components interact with the Arduino coding language. The presentation concludes with an introduction to the basic principles of circuit building that will be necessary to wire the components together.



2. Build Session (45 Minutes)

Students begin by customising their Alarm Bot's chassis from a wide variety of colour choices. They then assemble and wire all the components together using our building manual , with support from our team throughout the process.

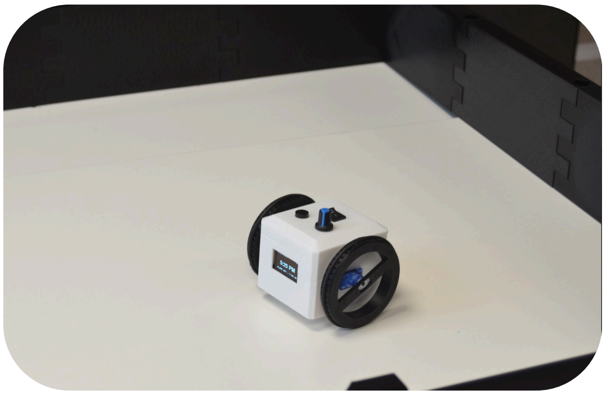


3. Coding Demonstration (15 Minutes)

A live demo session guides students through the process of setting up Arduino IDE and coding their buzzer sequences and their motion paths. To reduce complexity we will provide students code templates and only teach them a small portion of the bot's code.

4. Programming and Testing (30 Minutes)

Students apply their knowledge learnt during the coding demonstration to code their very own buzzer sequence and motion paths. They will also have the opportunity to test these codes with the support of our own workshop team and within a provided arena.



5. Trivia and Q&A Session (15 Minutes)

To conclude the workshop, our team will host an interactive trivia session testing knowledge taught throughout the workshop. Students will have the opportunity to win our 3D-printed prizes. If time is forgiving, we will also be happy to answer any questions in regards to the burgeoning STEM industry and tips to making the most of the university experience.

LEARNING OUTCOMES

All of our workshops are designed with NESA's ciruculum in mind. Though this workshop is specifically designed for the following subjects we are more than happy to host workshops for extracirrucular activities (i.e. robotics clubs).

SUBJECT	RELEVANT LEARNING OUTCOMES
Mandatory Technology (Years 7-8)	• TE4-MSC-01 - explains how materials, systems and components contribute to solutions • TE4-SAF-01 - selects and safely uses tools, materials, technologies and processes • TE4-DIG-02 - uses data and digital systems to code, design and produce projects
iSTEM Elective (Years 9-10)	• ST5-3 - applies engineering design processes to address real-world STEM-based problems • ST5-4 - works independently and collaboratively to produce practical solutions to real-world scenarios • ST5-5 - analyses a range of contexts and applies STEM principles and processes
Computing Technology Elective (Years 9-10)	• CT5-OPL-01 - designs, produces and evaluates algorithms and implements them in a general-purpose and/or object-oriented programming language • CT5-THI-01 - applies computational, design and systems thinking to the development of computing solutions

NEED TO KNOWS

Workshop Details

Price: $30 per student ($25 for low SES schools)
Duration: 2 hours
Workshop Type: Incursion
Risk Assessment: Google Drive Link

Requirements for Schools

Althought we provide almost all supplies and a testing arena we do ask that all prospective bookings meet the following requirements:

- There is an available Smartboard/TV screen that can be connected to via a HDMI port or by Microsoft's Screensharing
- All students have readily access to computers/laptops with **Arduino IDE pre-installed**
- A 1 metre by 1 metre space is available for our testing arena
- Constant supervision by a teacher during the workshop (we will roster enough instructors for an instructor-student ratio of approximately 1 : 5)

Public Liability Insurance

Public Liability Limit: $20,000,000
Date of Issue: 29/02/2026
Period of Cover: 28/02/2026 to 28/02/2027
Insurer Name: Chubb Insurance Australia Limited
Insurer ABN: 23 001 642 020
AFS Licence No. : 239687

How to Book this Workshop or Contact Us

If you wish to book our Alarm Bot Workshop please click on the "Book Now" button at the top right of our website **projectbeacon.org.au** and we will get back to you via email within a business day. If you have any further enquiries feel free to email us at **support@projectbeacon.org.au**. Hope to hear from you soon!